

# VoteHubPH Project Documentation

## Project Overview

VoteHubPH is a comprehensive voting and candidate information platform designed specifically for the Philippines. The platform enables Filipino voters to access detailed information about candidates running for various government positions, from national to barangay levels. The project aims to empower Filipino voters with accessible, accurate, and comprehensive information about candidates, enabling informed democratic participation in Philippine elections.

## Project Objectives

### Primary Objectives

- **Transparency:** Provide detailed candidate information including platforms, achievements, and education
- **Accessibility:** Enable voters to easily browse and filter candidates by location and position
- **Community Engagement:** Allow users to vote, comment, and discuss candidates
- **Multi-language Support:** Support for English and Filipino languages
- **Comprehensive Coverage:** Cover all government levels from National to Barangay

### Secondary Objectives

- **Performance:** Optimize for speed and efficiency with caching strategies
- **Scalability:** Build for future growth and expansion
- **Security:** Implement robust authentication and data protection
- **Responsive Design:** Ensure optimal user experience across all devices
- **SEO Optimization:** Implement features that help content reach a wider audience

## Technology Stack

### Frontend Technologies

- **Next.js 14:** React framework with App Router and SSR capabilities
- **React 19:** UI library for interactive components
- **TypeScript 5:** Type-safe JavaScript development

- **Tailwind CSS 4:** Utility-first CSS framework
- **NextAuth.js 4:** Authentication and session management
- **Prisma 6:** Database ORM for type-safe queries
- **Radix UI:** Accessible component primitives
- **React Hook Form 7:** Form validation and management
- **Zod 3:** Schema validation library
- **Lucide React:** Icon library for UI elements

### **Backend Technologies**

- **PHP 8.2+:** Server-side scripting language
- **Laravel 11:** Full-featured PHP framework
- **Laravel Sanctum 4:** API authentication
- **PostgreSQL 15+:** Relational database via Neon
- **Cloudinary SDK:** Image management and CDN
- **PHPUnit 11:** Testing framework

### **Infrastructure & Services**

- **Vercel:** Frontend hosting and deployment
- **Railway/Render:** Backend hosting
- **Neon:** Serverless PostgreSQL database
- **Cloudinary:** Image CDN and transformation
- **Docker:** Application containerization
- **GitHub:** Version control and CI/CD

### **Project Features**

#### **Core Features**

##### **1. User Authentication System**

- Email OTP registration with 6-digit codes
- Google OAuth 2.0 integration

- JWT token-based API authentication
- Profile management and completion
- Password recovery and account deletion

## **2. Content Management**

- Create and edit candidate posts with rich data
- Image upload and management via Cloudinary
- Candidate information fields: name, position, biography, platform, education, achievements
- Post status workflow: Draft → Pending → Approved/Rejected
- Admin moderation with rejection notes

## **3. Candidate Discovery**

- Multi-level filtering: National, Local, Barangay
- Location-based search: Region → Province → City → Barangay
- Position-based filtering
- Party list directory browsing
- Search functionality across candidates

## **4. User Interaction**

- Voting system with optional anonymity
- Threaded comment system with replies
- Comment like/unlike functionality
- Real-time vote counting
- User notifications

## **5. Dashboard & Analytics**

- Personal user dashboard
- Post management interface
- Vote and comment history

- User statistics and engagement metrics
- Settings and preferences management

## 6. Admin Features

- Post moderation dashboard
- Admin approval/rejection workflow
- Party list management
- Member management for party lists
- Statistics and engagement analytics
- Admin authentication system

## Design Features

- **Dual Theme Support:** Light and dark mode toggle
- **Responsive Layout:** Optimized for desktop, tablet, and mobile devices
- **Accessibility:** WCAG 2.1 compliant design elements
- **Performance First:** Optimized for speed with caching strategies
- **User-Centric Design:** Prioritizing user experience throughout

## Project Structure

### Page Components

#### 1. Public Pages

- Home/Landing page with platform overview
- About page with project information
- Browse candidates page with filters
- Individual candidate detail pages
- Party list directory
- Election information page
- User authentication pages (login/register)

#### 2. User Dashboard

- Personal dashboard overview
- My Posts management interface
- Analytics and statistics page
- Settings and preferences
- Notifications page
- Profile management

### 3. Community Features

- Community page
- User profile pages
- Search and discovery tools
- Party list browsing
- Candidate voting interface

### Target Audience

#### Primary Users

- **Filipino Voters:** Citizens seeking candidate information and voting
- **Content Contributors:** Users submitting candidate information
- **Civic-minded Filipinos:** Community members interested in democratic participation

#### Secondary Users

- **Tourists:** Visitors seeking information about Philippine politics
- **Diaspora Community:** Filipinos living abroad staying connected
- **Researchers:** Academics studying Philippine elections and governance

### Development Approach

#### Design Philosophy

- **User-Centric Design:** Prioritizing user experience and accessibility
- **Performance First:** Optimizing for speed and efficiency
- **Scalability:** Building for future growth and expansion

- **Security Focus:** Protecting user data and ensuring safe interactions
- **Responsive Design:** Mobile-first development approach

### Development Methodology

- **Component-Based Architecture:** Reusable UI components for consistency
- **Accessibility-First:** Ensuring inclusive design from the start
- **Progressive Enhancement:** Building core functionality first, then adding enhancements
- **Test-Driven Development:** Comprehensive testing for reliability
- **Continuous Integration:** Automated testing and deployment

### Database Design

#### Key Models

- **User & Authentication:** User accounts, OAuth sessions, OTP verification
- **Content:** Posts, comments, votes, comment likes
- **Party Lists:** Party information and member details
- **Location:** Region, Province, City, Barangay hierarchies

#### Key Features

- 13 Eloquent models with proper relationships
- PostgreSQL database via Neon
- Cascading deletes and foreign key constraints
- Indexes on frequently queried columns
- Unique constraints on votes and likes

### API Architecture

#### Authentication Endpoints

- POST /api/auth/send-otp - Send OTP code
- POST /api/auth/verify-otp - Verify and register user
- POST /api/auth/google - Google OAuth callback
- POST /api/admin/login - Admin authentication

## **User Endpoints**

- GET /api/user/profile - Get user profile and activity
- POST /api/user/complete-profile - Complete OAuth profile
- PUT /api/user/update - Update user information
- DELETE /api/user/delete-account - Delete user account

## **Post Endpoints**

- POST /api/posts - Create candidate post
- GET /api/posts/approved - Get all approved posts
- GET /api/posts/{id} - Get post details
- GET /api/posts/my-posts - Get user's posts
- PUT /api/posts/{id} - Update post

## **Admin Endpoints**

- GET /api/admin/posts - Get all posts with status
- GET /api/admin/posts/pending - Get pending posts
- POST /api/admin/posts/{id}/approve - Approve post
- POST /api/admin/posts/{id}/reject - Reject post

## **Voting & Comments**

- POST /api/posts/{postId}/vote - Submit vote
- GET /api/posts/{postId}/vote-status - Check user's vote
- GET /api/posts/{postId}/comments - Get comments
- POST /api/posts/{postId}/comments - Post comment
- POST /api/comments/{commentId}/like - Like comment

## **Location Endpoints**

- GET /api/locations/regions - Get all regions
- GET /api/locations/provinces - Get provinces
- GET /api/locations/cities - Get cities

- GET /api/locations/barangays - Get barangays

### Party List Endpoints

- GET /api/partylists - Get active party lists
- GET /api/partylists/{id} - Get party details
- GET /api/partylists/search - Search party lists
- POST /api/admin/partylists - Create party list
- POST /api/admin/partylists/{id}/members - Add members

### Security & Authentication

#### Security Measures

- **Password Hashing:** bcrypt with salt rounds
- **OTP Rate Limiting:** 24-hour tracking with increasing cooldowns
- **CORS Configuration:** Restricted to specific origins
- **CSRF Protection:** Laravel CSRF tokens
- **JWT Validation:** Custom middleware verification
- **HTTPS Only:** All production traffic encrypted
- **Session Management:** Secure database-backed sessions

#### Authentication Strategies

- **Email OTP:** Secure signup verification
- **Google OAuth:** Third-party authentication
- **JWT Tokens:** API token validation
- **Admin Sessions:** Separate admin authentication layer

### Configuration

#### Environment Setup

Backend requires:

- Database credentials (PostgreSQL via Neon)
- Email service configuration (Gmail SMTP)



- Cloudinary credentials
- Laravel application key

Frontend requires:

- NextAuth configuration
- Google OAuth credentials
- API endpoint configuration
- Cloudinary upload presets

### **Key Dependencies**

Backend: Laravel framework, PostgreSQL driver, Cloudinary SDK, PHPUnit

Frontend: Next.js, React, NextAuth.js, Prisma, Tailwind CSS

### **Business Logic & Services**

#### **OTP Management**

- 6-digit codes with 5-minute expiration
- Rate limiting: 1-minute cooldown for first 2 attempts
- 3-minute cooldown for subsequent attempts
- 24-hour tracking per email address

#### **Post Workflow**

1. User submits candidate post
2. System sets status to "Pending"
3. Admin reviews submission
4. Admin action: Approve → Visible, or Reject → Removed with notes
5. User can edit pending posts
6. Approved posts appear in browse and search

#### **Voting Rules**

- One vote per user per candidate
- Optional anonymous voting

- Votes can be toggled (remove vote)
- Real-time vote count updates
- Vote ownership tracking

### **Comment Threading**

- Top-level comments have no parent
- Nested replies reference parent comment
- Anonymous comments show "Anonymous" author
- Like counts tracked per comment
- Delete cascade for removed comments

### **Caching Strategy**

- Location data: 1-hour cache
- Party lists: 30-minute cache
- Statistics: 15-minute aggregate cache
- Database-backed caching with fallback

### **Deployment Guide**

#### **Prerequisites**

- Node.js 18+ installed
- PHP 8.2+ with Composer
- PostgreSQL database access (Neon account)
- Cloudinary account
- Google Cloud project for OAuth
- Vercel account for frontend
- Railway or Render account for backend

#### **Backend Deployment Steps**

1. Prepare environment with .env configuration
2. Install dependencies: composer install

3. Generate application key
4. Run migrations: `php artisan migrate`
5. Deploy to Railway with connected GitHub repo

### **Frontend Deployment Steps**

1. Connect GitHub repository to Vercel
2. Select frontend directory as root
3. Configure environment variables in dashboard
4. Auto-deploys on push to main branch

### **Database Setup**

1. Create Neon PostgreSQL project
2. Copy connection string
3. Configure `DATABASE_URL`
4. Run Prisma migrations: `npx prisma migrate deploy`

### **Cloudinary Configuration**

1. Create Cloudinary account
2. Create unsigned upload preset: "votehubph\_unsigned"
3. Add credentials to environment variables

### **Google OAuth Setup**

1. Create Google Cloud project
2. Enable Google+ API
3. Create OAuth 2.0 credentials
4. Add redirect URI: `https://your-domain/api/auth/callback/google`
5. Configure client ID and secret

### **Testing**

#### **Backend Testing**

- PHPUnit test suite for unit and feature tests

- Test configuration with in-memory SQLite database
- Run tests: php artisan test
- Coverage reports available

### **Frontend Testing**

- ESLint for code quality
- TypeScript validation
- Build verification
- Manual testing checklist provided

### **Manual Testing Checklist**

- ✓ User registration (Email OTP)
- ✓ User registration (Google OAuth)
- ✓ Profile completion
- ✓ Candidate browsing with filters
- ✓ Location-based filtering
- ✓ Post creation and editing
- ✓ Voting functionality
- ✓ Comment posting and replies
- ✓ Admin moderation workflow
- ✓ Party list browsing
- ✓ Theme switching
- ✓ Mobile responsiveness
- ✓ API endpoint validation

## Government Levels & Positions

**National Level:** President, Vice President, Senator, Party-list Representative, District Representative

**Local Level:** Governor, Vice Governor, Board Member, Mayor, Vice Mayor, Councilor

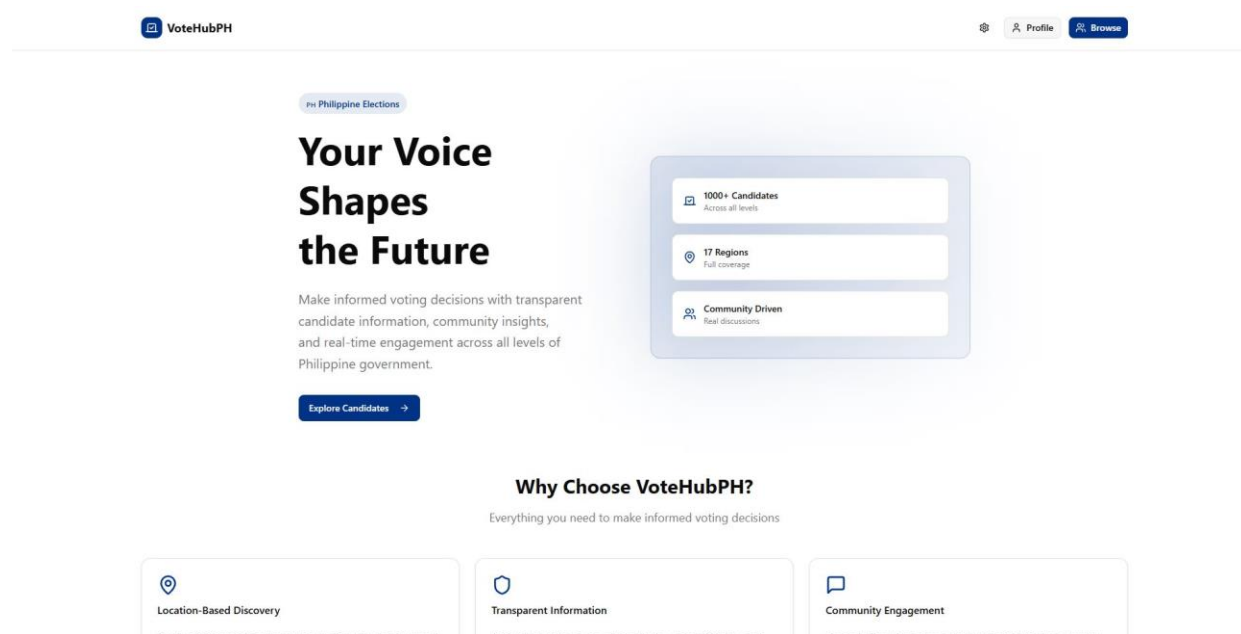
**Barangay Level:** Barangay Captain, Barangay Kagawad, SK Chairman

## Party List Sectors

Youth, Women, Labor, Indigenous Peoples, Farmers, Fisherfolk, Senior Citizens, Persons with Disabilities, Overseas Filipino Workers, Urban Poor, Cooperative, Professional

## Page Overview

### User Side



### Landing Page

← Back

Refreshing...

Create Post

LocalNationalParty-List

CandidatesElections

Select Your Location

Choose your region, city, and barangay to see relevant candidates

Auto-detect

Region

National Capital Region (NCR)

City/Municipality

Manila

District

Select district

Search candidates by name or platform...

Local Candidates

3 candidates found

Create Candidate Post

Ryu Nakayama

Running for SK Chairperson

KEY ACHIEVEMENTS:

✓ Embracing and guidance for the youth

✓ Sports for the youth and adults

✓ Medicine giving

1 votes

3 comments

View Profile

Honey Lacuna

Running for Mayor

Lacuna babies

ABOUT

Tapat at Totoo

0 votes

0 comments

View Profile

Isko Moreno

Running for Mayor

Independent

KEY ACHIEVEMENTS

✓ mayor

0 votes

0 comments

View Profile

## Candidates Page

← Back

Create Post

LocalNationalParty-List

CandidatesElections

Election Schedule

Upcoming elections and voting dates

BARMMParliamentary Elections 2026

May 11, 2026

First regular Bangsamoro Parliamentary Elections under the postponed schedule. Elections to be held every three years thereafter.

Positions:

BARMMParliament Members (80 seats)

Local

Barangay and Sangguniang Kabataan Elections 2026

November 2, 2026

Nationwide Barangay and SK Elections. All 42,000+ barangays across the Philippines will elect their local leaders. Postponed from December 2025 to November 2026.

Positions:

Barangay Chairman

Barangay Kagawad (7 seats per barangay)

SK Chairman

+1 more

Local

2028 Philippine Presidential Elections

May 8, 2028

Presidential, Vice Presidential, Senatorial, and Congressional Elections. Filipino citizens will elect the next President and Vice President for a 6-year term.

Positions:

President

Vice President

Senators (12 seats)

+2 more

National

## Election Schedule Page

[← Back](#)

RefreshCreate Post

LocalNationalParty-List

CandidatesElections

Search party-list groups by name, acronym, or sector...

Party-List Groups3 groups found

Create Candidate Post

Independent

1 memberView Details →

Lacuna babies

1 memberView Details →

party party raw

1 memberView Details →

Party List Directory

[← Back to Party-List](#)

Independent

1 member

Members



Isko Moreno

Position #1Local (City/Municipality)Mayor

Key Achievements:

- mayor

View Profile

Party List Content

[← Back to Browse](#)

My ProfileManage your civic engagement



Andrei Condino

andreicondino2@gmail.com

Joined Dec 7, 2025

Edit Profile

My Posts

My Votes

My Comments

Voted Candidates

Candidates you've voted for



Isko Moreno

Mayor • Local (City/Municipality)

AnonymousDec 7, 2025

View Profile

User Profile



Isko Moreno

Running for Mayor

Independent

Local (City/Municipality)

0 votes

3 comments

Support This Candidate

Vote for Isko

☒ Vote anonymously

This is a community poll to gauge voter sentiment. Your vote will be anonymous and help others understand public opinion.

About

hello

Platform & Advocacy

ewan

Key Achievements

✓ mayor

Community Discussion

What voters are saying about this candidate

Share your thoughts about this candidate...

☒ Post anonymously

Post Comment

RECENT COMMENTS (2)

?

Anonymous

12/1/2025

Very Nice

0

Reply

A

Andrei Condino

12/1/2025

Holy

0

Reply

A

Andrei Condino

12/1/2025

Good bro

0

Reply

About This Profile

Candidate profiles are created and maintained by community members. All information should be verified through official sources.

## Candidate Profile



### Create a Post

Share your profile and platform as a candidate

#### Your Name \*

Mark Andrei Condino

#### Profile Photo \*

Upload a professional photo of yourself



#### Party / Partylist (Optional)

POGI

#### Level \*

National

#### Position \*

President

#### Your Bio \*

I am a Good Politician

22/500

#### Education

College

TIP

+

3/100

## Candidate Creation Page

## Admin Side



**VoteHubPH Admin**  
Post Moderation Dashboard

[Refresh](#)

[View Main Site →](#)

6

Total Posts

0

Pending Review

4

Approved

2

Rejected

[All Posts](#)

[Pending](#)

[Approved](#)

[Rejected](#)



**decline**

[Rejected](#)

[RYU NAKAYAMA](#) (mrnakayama@tip.edu.ph)  
President • National • [decline](#)

Submitted  
11/12/2025  
2:11:58 PM

Bio

a

Education

• Elementary - a

#### Admin Notes

Post rejected by admin



**LBERON**

[Rejected](#)

[MICHAEL ANGELO TORRES](#) (mmatorres01@tip.edu.ph)  
Barangay Captain • Barangay • [LAKAS BUTO](#)

Submitted  
11/12/2025  
2:10:45 PM